

Possible Mutations Information Sheet

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1 Overview of Target Mutations

Only target regions within nine genes are considered in this analysis. These genes were identified in:

Moloney-Geany, K., Black, M.A., Day, R. et al. A set cover algorithm identifies minimal circulating tumour DNA sequencing targets for colorectal cancer detection. Sci Rep 15, 40949 (2025). <https://doi.org/10.1038/s41598-025-24719-9>

The target genes include oncogenes and tumour suppressors with differing mutation criteria, summarised below.

1.1 Gene-specific mutation definitions

Oncogenes (KRAS, BRAF, NRAS, CTNNB1, PIK3CA): Only mutations classified as pathogenic in COSMIC or ClinVar are considered positive.

ACVR2A and RNF43: Each gene contains a single targeted hotspot, capturing approximately 85% and 90% of somatic mutations in colorectal cancer respectively. Only these hotspot variants are considered positive.

APC (tumour suppressor): Over 99% of driver mutations are truncating or splice-site variants. Therefore, only truncating and splicing mutations are considered positive. Missense and other in-frame variants are excluded.

TP53 (dual role gene): TP53 includes missense, truncating, splice-site, and in-frame INDEL mutations. Truncating and splice-site variants are always considered positive. Missense and in-frame INDELs are only considered positive if previously reported as pathogenic in COSMIC or ClinVar.

2 Identification of SNP (Substitution) Mutations

2.1 COSMIC database search strategy

For each gene, possible substitution mutations were identified using the COSMIC database. Each gene was queried individually using the advanced filters.

The following filters were applied:

- Assay Type: All
- Somatic Status: Confirmed Somatic
- Sample Type: Tumour only
- Mutation Impact: Pathogenic
- Location: Large Intestine
- Type: Missense or Nonsense

Splice-site mutations are not comprehensively represented in COSMIC, as the database focuses on coding regions.

2.2 Splice-site mutation sources

Splice-site variants were sourced from cBioPortal and multiple colorectal cancer datasets listed below.

Cancer Type	Cohort/Source	Journal
Colorectal Adenocarcinoma	DFCI	Cell Reports 2016
Colorectal Adenocarcinoma	Genentech	Nature 2012
Colorectal Adenocarcinoma	TCGA	PanCancer Atlas
Colorectal Adenocarcinoma Triplets	MSK	Genome Biology 2014
Colorectal Cancer	CAS Shanghai	Cancer Cell 2020
Colorectal Cancer	HTAN MSK	Nature 2025
Colorectal Cancer	MSK	Cancer Discovery 2022
Colorectal Cancer	MSK	Gastroenterology 2020
Colorectal Cancer	MSK	JCO Precision Oncology 2022
Colorectal Cancer	ChangKang Project SYSUCC	Nature Communications 2022
Rectal Cancer	MSK	Nature Medicine 2019

Table 1: Colorectal cancer datasets used for splice-site variant identification

2.3 Substitution mutation output format

All substitution variants were compiled into a CSV file with the following structure:

Column #	Field	Example Value
1	GENE_NAME	CTNNB1
2	Mutation CDS	c.121A>G
3	Mutation AA	p.T41A
4	GENOMIC_WT_ALLELE_SEQ	A
5	GENOMIC_MUT_ALLELE_SEQ	G
6	Mutation Description AA	Substitution - Missense
7	chrom	NC_000003.12
8	pos	41224633

Table 2: CSV format for substitution (SNP) mutations

3 Identification of INDEL Mutations

INDEL mutations were identified using the same cBioPortal datasets listed previously for splice-site variant discovery. COSMIC was not used because, despite having information on INDEls, the use of 'pathogenic' and 'confirmed somatic' filtered resulted in no results.

Each dataset was queried by gene, and all variants for the target genes were downloaded as TSV files. These were processed using an R Markdown pipeline (cBioPortal.INDEL.Processing.Rmd) to filter and classify variants.

3.1 INDEL mutation output format

The final INDEL dataset was compiled into a CSV file with an expanded format to account for variable-length genomic changes:

Column #	Field	Example Value
1	GENE_NAME	CTNNB1
2	Mutation CDS	c.121A>G
3	Mutation AA	p.T41A
4	GENOMIC_WT_ALLELE_SEQ	A
5	GENOMIC_MUT_ALLELE_SEQ	G
6	Mutation Description AA	Substitution - Missense
7	chrom	NC_000003.12
8	Start.Pos	41224633
9	End.Pos	41224634

Table 3: CSV format for INDEL mutations (including end coordinates)